Massa Corporis et Densitas Architecturae

Length Is Not Size: Re-estimating Body Mass in the Largest Living Cetaceans and Their Mesozoic Marine Counterparts

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Abstract

Published body masses for the largest living cetaceans rest on very few weighed individuals, and the most often repeated maxima are either unweighed figures or values taken from animals well short of the longest recorded lengths. We re-estimate the mass of large killer whales (*Orcinus orca*) and Antarctic blue whales (*Balaenoptera musculus intermedia*) by isometric scaling from explicitly stated weighed anchors, set the results beside independent estimates, and compare them with mass estimates for Mesozoic marine reptiles of similar length. From a reported Soviet-era anchor of 8.6 m and 9.9 t, a killer whale of 9.45–9.70 m notch-to-fluke length is predicted at 13.1–14.2 t under strict isometry, and at 14.4–17.0 t if its girth exceeds the anchor's by 10–20%, as photographs of the transient male CA165 suggest. Regressions fitted to adult Antarctic whales give 13.4–15.1 t over the same lengths, so the strict-isometry values are what adult data predict; the open question is length, since the largest males in measured samples are 9.0–9.45 m and CA165 has not been measured. From the 27.6 m blue whale weighed in 1947, whose maternal mass we take as about 190 t after removing an estimated gestational load, isometry predicts about 244 t at 30 m and 325 t at 33 m, against 184–205 t and 252–273 t from a published regression; the anchor was evidently in heavy condition. On every reading a 33 m blue whale weighed 250 t or more. To compare body plans independently of length we use a fullness coefficient $k = M/L^3$. In each of four length tiers the cetacean exceeds the reptile it is compared with, by a factor of 1.3–2.6 at range midpoints. Large killer whales therefore match or exceed all but a few mosasaurid species in mass, and fall in the same 7–17 t band as *Tyrannosaurus rex*, *Spinosaurus aegyptiacus* and *Mosasaurus hoffmannii*. Every input is tagged by evidence status, so that each conclusion can be traced to what it rests on.

Keywords: *Orcinus orca*; *Balaenoptera musculus*; body mass; isometric scaling; sampling bias; Mosasauridae; Ichthyosauria; CA165

Note on this version. This draft supersedes the earlier Word versions. The 17-point killer whale mass table and the power-law fit derived from it ($b = 3.71$) have been removed, because those points were illustrative rather than recorded measurements; all killer whale estimates now rest on the single stated anchor. The giant ichthyosaur previously labelled "Austrian" is the Aust Cliff material from Gloucestershire, UK. Independent estimates for both whales are now shown beside the isometric values. A previous build quoted a single mixed-age model value (11.5 t at 9.5 m) as the independent estimate for killer whales; that was a misreading, and the adult regressions and the measured length samples are now tabulated instead. Every input carries an evidence tag (§2.1).

1 Introduction

Body length is the size measure most often quoted for large marine vertebrates, living or extinct, because it is the easiest to measure on a carcass or to reconstruct from a skeleton. Mass, however, is what matters for most questions of physiology, locomotion and ecology, and it does not follow

from length alone. It depends on how much of the length is occupied by a full, deep trunk rather than by a tapering tail.

For living cetaceans, published masses rest on remarkably few weighed animals. No killer whale longer than about 9 m appears ever to have been weighed; the widely quoted figure of "more than 10 t" for a 9.8 m male (Guinness World Records, 2019) is given without any record of a weighing. The most-cited maximum for the blue whale, about 190–200 t, comes from a 27.6 m female, almost 6 m shorter than the longest recorded.

For extinct marine reptiles the bias tends to run the other way: popular accounts often represent a species by its largest and most fragmentary extrapolation. When a modern population mean is set beside a fossil maximum, and both are expressed as length, the comparison exaggerates the size gap between Mesozoic and modern marine predators.

1.1 Aims

We (i) state single, traceable mass anchors for *O. orca* and *B. m. intermedia*; (ii) extrapolate them by isometric scaling, with an explicit girth correction, to the largest documented individuals; (iii) compare cetaceans and marine reptiles of similar length using a length-independent fullness coefficient; and (iv) place the results beside current mass estimates for the largest Mesozoic predators.

2 Materials and methods

2.1 Evidence status

Each input value carries one of four tags: [P], published in a peer-reviewed primary source cited here; [R], reported in a secondary or grey-literature source, or attributed to a primary source that has not yet been checked against the original table; [A], the author's assessment, such as a length read from field photographs or a percentile assignment; and [T], a theoretical value produced by the scaling model of this paper. Table 1 lists the inputs.

2.2 Length conventions

Cetacean lengths are notch-to-fluke lengths, from the tip of the snout to the notch between the flukes, the standard cetacean measurement. Length to the fluke tips exceeds this by about 0.20–0.25 m in a large killer whale, but the fluke lobes are thin connective tissue and add negligible mass, so the choice of convention does not materially affect mass scaling. Fossil lengths are reconstructed total lengths.

2.3 Isometric scaling and the fullness coefficient

Under geometric similarity, mass scales with the cube of length. A target individual of length L is estimated from an anchor of length L_0 and mass M_0 as

$$M = M_0 \left(\frac{L}{L_0} \right)^3 G \tag{1}$$

where G is a girth factor expressing how much fuller the target's cross-sections are than the anchor's; $G = 1$ is strict isometry. To compare animals of different length and body plan we use the fullness coefficient

Table 1. Input values and their evidence status.

Quantity	Value	Status	Source or basis
O. orca anchor, notch-to-fluke length	8.6 m	[R]	Soviet Antarctic whaling data, attributed to Mikhalev et al. (1981)
O. orca anchor, mass (fluid loss compensated)	9.9 t	[R]	as above
O. orca conservative anchor mass	9.0 t	[A]	author's lower reading of the same record
CA165, field length estimate	9.1–10.0 m (30–33 ft)	[R]	Killer Whale Wiki (n.d.)
CA165, adopted notch-to-fluke length	9.45–9.70 m	[A]	aerial and lateral footage
CA165, girth factor G	1.10–1.20	[A]	visual assessment of footage
9.8 m male listed at "more than 10 t"	–	[R]	Guinness World Records (2019); no weighing recorded
B. musculus anchor	27.6 m; 190 t piecemeal, ≈199 t intact	[R]	Wood (1982)
Gestational load (fetus + amniotic fluid)	6–7 t + 2–3 t	[A]	late-term estimate
B. musculus, regression and volumetric masses	25 m: 101–119 t; 30 m: 184–205 t; 33 m: 252–273 t	[R]	Motani and Pyenson (2024), values as summarised by Wikipedia (2026)
O. orca, length–mass regressions	Table 3	[R]	Cetology Hub (2024)
O. orca, measured male lengths	Table 4	[P]/[R]	see table
B. musculus, longest recorded length	33.58 m	[R]	Wood (1982)
Mean length, mature Antarctic females	25.4–26.3 m	[R]	Wikipedia (2026)
Percentile positions, Antarctic females	Table 5	[A]	author's assignment
Fossil lengths and masses	Tables 6, 7	[P]/[R]/[A]	see tables

$$k = \frac{M}{L^3} \tag{2}$$

which is constant along any isometric series. A higher k means more mass per cube of length: a deeper trunk, a heavier skeleton, a thicker blubber layer, or a shorter tail relative to the trunk. We report k in units of $10^{-3}\,\mathrm{t\,m^{-3}}$.

2.4 Killer whale anchor and CA165

The anchor is a male of 8.6 m notch-to-fluke length, weighed at 9.9 t with compensation for fluid loss during Soviet Antarctic whaling operations. It is attributed to the data summarised by Mikhalev et al. (1981), and the figure should be checked against the original tables before publication [R].

It lies within 3% of the regression later published for Antarctic whales (§3.1). As a conservative alternative we also use 9.0 t at the same length [A].

CA165 ("Lonesome George") is an adult male Bigg's (transient) killer whale photographed off California, with field length estimates of 30–33 ft (9.1–10.0 m) and a dorsal fin of about 2 m (Killer Whale Wiki, n.d.). From aerial and lateral footage we adopt a notch-to-fluke length of 9.45 m (conservative) to 9.70 m (upper) [A]. Observers repeatedly describe him as unusually round in the body, so we apply $G = 1.10$–1.20 [A].

2.5 Blue whale anchor and gestational correction

The anchor is the female of 27.6 m taken in the Southern Ocean on 20 March 1947, weighed piecemeal at 190 t with an estimated intact mass of about 199 t (Wood, 1982). She is reported to have been pregnant [R]. A late-term blue whale fetus weighs roughly 6–7 t and its amniotic fluid perhaps 2–3 t [A], so we take her maternal mass as about 190 t, that is 199 t less a gestational load of about 9 t.

On the length distribution of sexually mature Antarctic females the author places 27.6 m at the 45th–65th percentile, lengths above 30 m at the 90th–95th, and 32–33 m at the 99th [A]. These assignments have not yet been checked against catch-length data, and the summary statistics available suggest that they are too low (§4.5). The longest recorded blue whale is a female of 33.58 m (107 Norwegian *fot*) (Wood, 1982); the longest measured under scientific conditions is 29.9 m (Wikipedia, 2026). Lengths of 35–37 m are treated only as theoretical upper bounds [T].

2.6 Cross-taxon tiers

We compare cetaceans and marine reptiles in four length tiers: I, 7–8 m (*Globicephala melas* and *Tylosaurus proriger*); II, 9–12 m (*O. orca*, *Thalassotitan atrox* and *M. hoffmannii*); III, about 16.5 m (*Physeter macrocephalus* and *Himalayasaurus tibetensis*); and IV, 20–36 m (*B. musculus* and large shastasaurid ichthyosaurs). Where a source gives a range we carry the range through, and we summarise each taxon by the k of its range midpoints. Fossil masses are themselves reconstructions, so the comparison is between best current estimates, not between weighed animals on both sides.

3 Results

3.1 Killer whale: CA165

Table 2 and Figure 1 give the estimates. Under strict isometry the anchor predicts 13.1 t at 9.45 m and 14.2 t at 9.70 m. With the girth factor the range widens to 14.4–17.0 t; the value of 16.7 t used in earlier drafts corresponds to 9.70 m with $G \approx 1.18$. Even the most conservative combination considered (9.0 t anchor, 9.45 m, $G = 1$) gives 11.9 t, well above the "about 10 t" commonly cited for the largest males. Figure 2 shows how the estimate moves with length and girth.

These values agree with regressions fitted to adult whales (Table 3). The relation of Mikhalev (2019) for 95 weighed Antarctic killer whales has an exponent of 2.97, almost exactly isometric, and gives 10.1 t at 8.6 m: the anchor is an ordinary Antarctic male, not a heavy one. At 9.45–9.70 m that relation gives 13.4–14.5 t, and a relation for South African type A whales and a compiled relation for whales longer than 4.5 m give up to 15.1 t [R]. Relations fitted mainly to young or mixed-age animals have exponents of 2.5–2.8 and give 9.7–12.2 t over the same lengths. The best known of them, that of Bigg and Wolman (1975), rests on live-captured animals, most of them juveniles (Noren, 2011), and should not be extrapolated to large bulls.

Table 2. Mass estimates for CA165 under alternative anchors, lengths and girth factors.

Scenario	Anchor	L (m)	G	M (t)	Status
Strict isometry, conservative length	8.6 m, 9.9 t	9.45	1.00	13.1	[T]
Strict isometry, upper length	8.6 m, 9.9 t	9.70	1.00	14.2	[T]
Girth-corrected, conservative length	8.6 m, 9.9 t	9.45	1.10–1.20	14.4–15.8	[T]
Girth-corrected, upper length	8.6 m, 9.9 t	9.70	1.10–1.20	15.6–17.0	[T]
Conservative anchor mass	8.6 m, 9.0 t	9.45–9.70	1.00	11.9–12.9	[T]
Shorter anchor length	8.4 m, 9.9 t	9.45–9.70	1.00	14.1–15.2	[T]

The last row tests the possibility that the anchor's notch-to-fluke length was below 8.6 m: a shorter anchor of the same mass implies a fuller body and raises every estimate.

Table 3. Length–mass relations for killer whales, $M = aL^b$, evaluated at the anchor length and at the adopted lengths of CA165.

Relation	Basis	b	n	8.6 m (t)	9.45 m (t)	9.70 m (t)
Mikhalev (2019)	Antarctic catch	2.97	95	10.1	13.4	14.5
Best (2008)	type A, South Africa	3.28	7	10.0	13.6	14.9
Compiled	whales longer than 4.5 m	3.28	58	10.2	13.8	15.1
Bigg and Wolman (1975)	live capture, mostly juveniles	2.58	32	7.6	9.7	10.3
McClure (2024)	transients	2.54	13	8.6	10.9	11.6
Compiled	all sizes	2.77	102	8.7	11.3	12.2
This paper	isometric from anchor	3.00	1	9.9	13.1	14.2

Coefficients are those recalculated to metres and tonnes by Cetology Hub (2024) [R]; the originals of Mikhalev (2019), Best (2008) and McClure (2024) were not seen. The first three relations rest on adult or large whales, the next three mainly on young or mixed-age animals.

The strict-isometry values in Table 2 are therefore what adult data predict for an animal of this length. The girth-corrected values are the author's adjustment: 16.7 t at 9.70 m requires a whale 11–15% heavier than the adult regressions predict, a modest excess, but one that has not been measured for CA165.

3.2 The unweighed 9.8 m record

Handbooks give the largest killer whale as a male of 9.8 m weighing "more than 10 t" (Guinness World Records, 2019; Fisheries Agency of Japan, 2021); NAMMCO (n.d.) gives 10 m and 10 t. We could not trace either figure to a measurement. The primary Japanese data, 320 males reported in the coastal catch of 1948–1957, have a largest male of 31 ft (9.45 m) and contain no weights at all (Nishiwaki and Handa, 1958). A mass of 10 t at 9.8 m would imply $k = 10.6$, about two thirds of the anchor's 15.6. The adult regressions give 14.9–15.6 t at that length and the anchor curve 14.6 t. The relation of Bigg and Wolman (1975), extrapolated far beyond the young animals it was fitted to, gives 10.6 t, which may be where the "10 t" comes from [A].

3.3 How long are the largest males?

Mass at a given length is now the better constrained half of the problem; length is the weaker half. Table 4 and Figure 3 collect the samples in which males were measured. In each of the

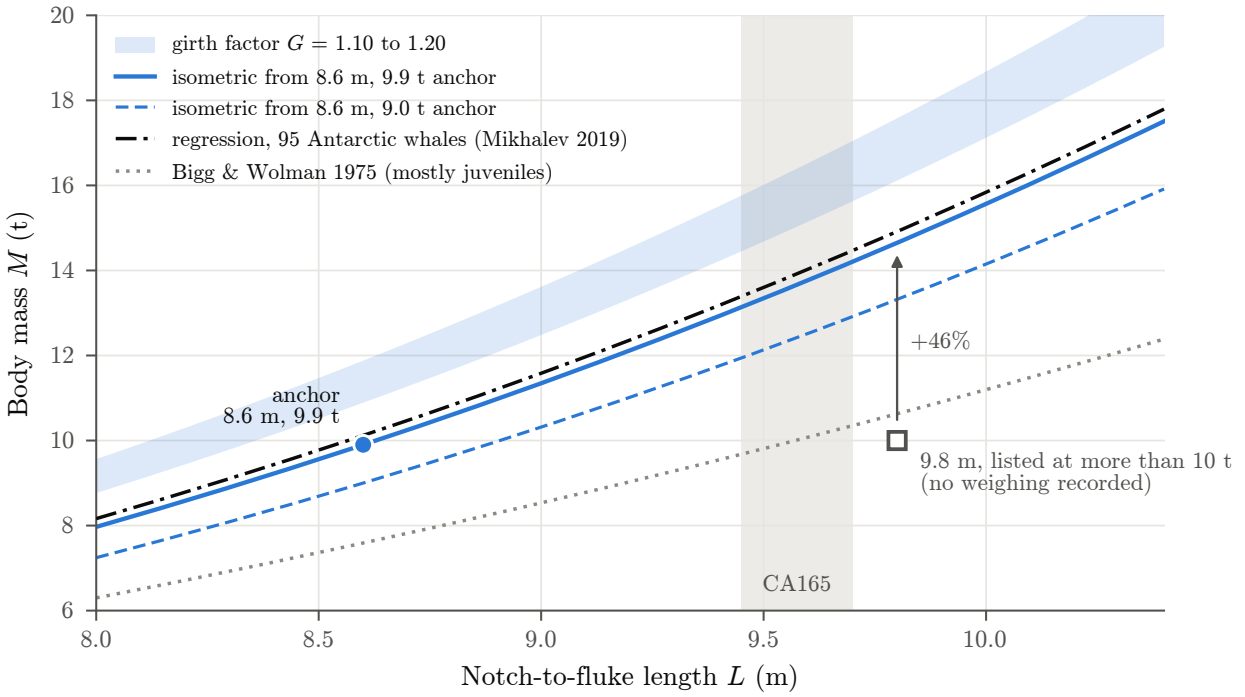


Figure 1. Isometric mass–length curves for *O. orca*. Solid line: scaling from the 8.6 m, 9.9 t anchor (circle). Dashed line: conservative 9.0 t anchor. Shaded band: girth factor $G = 1.10$–1.20 applied to the solid curve. Grey vertical band: adopted length of CA165. Square: the 9.8 m male listed at "more than 10 t", for which no weighing is recorded; the anchor curve predicts 14.6 t at that length. Dark dash-dotted line: regression for 95 weighed Antarctic whales (Mikhalev, 2019). Grey dotted line: the relation of Bigg and Wolman (1975), fitted mostly to juveniles.

two large catch samples the largest male is 9.0–9.45 m, against a mean of 6.5–7.3 m. The only photogrammetric sample of Bigg's killer whales, the ecotype of CA165, has 12 adult males of 6.3–8.3 m (Kotik et al., 2023); it comes from British Columbia and Washington and includes no California animals.

None of the individuals reputed to be 9 m or more has been measured. The 30–33 ft given for CA165 is a field estimate. For Old Thom, a solitary male of the Bay of Fundy, the press gives 25–30 ft (7.6–9.1 m) and about 8 t (Radio-Canada, n.d.). For John Coe, the best known male of the Scottish West Coast Community, no length has been published; his group is described as "about a metre bigger" than neighbouring populations (Hebridean Whale and Dolphin Trust, n.d.), and whales of his type reach 8.5 m, against 6.6 m for the commoner North Atlantic type (Foote et al., 2009).

The adopted length of CA165, 9.45–9.70 m, therefore equals or exceeds the largest male in every measured sample, and exceeds the largest photogrammetrically measured Bigg's male by more than a metre. Males of 9 m and more are real, but they are the extreme tail of large samples. A calibrated drone measurement of CA165 is the single most useful datum this paper lacks.

3.4 Antarctic blue whale

Table 5 lists masses at the lengths discussed, and Figure 4 plots them. Scaling the 190 t maternal anchor isometrically gives about 244 t at 30 m, 296–325 t at 32–33 m and 342 t at the longest recorded length of 33.58 m. These are the masses of whales as full-bodied as the 1947 female. Published

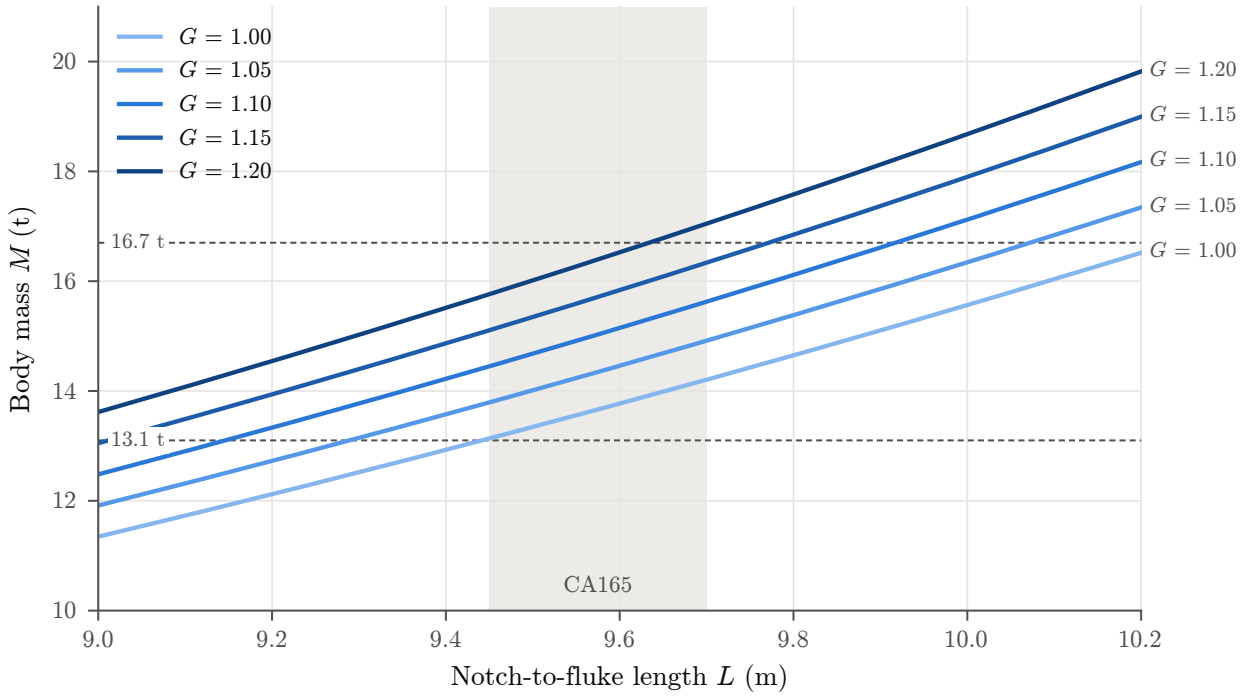


Figure 2. Sensitivity of the CA165 estimate to length and girth factor. Each line is equation (1) for one value of G. Dashed reference lines mark 13.1 t (strict isometry at the conservative length) and 16.7 t.

regression and volumetric estimates are lower: 101–119 t at 25 m, 184–205 t at 30 m and 252–273 t at 33 m (Motani and Pyenson, 2024) [R]. They correspond to $k = 6.5$–7.6, against 9.0 for the anchor, so the 1947 female was about 26% heavier than a typical whale of her length. She was pregnant, and she was taken on 20 March, at the end of the feeding season, when rorquals are at their heaviest.

The author's conservative bracket [A], at least 240 t at 30 m, 270–290 t at 33 m and 290–340 t as a theoretical bound for 35–37 m, lies between the two at 33 m and above the published value at 30 m. The three readings agree on the point that matters here: a 33 m blue whale weighed 250 t or more, so the familiar "200 t maximum" is the mass of one heavy 27.6 m animal, not the limit of the species.

3.5 Fullness across tiers

Table 6 gives lengths, masses and fullness coefficients; Figure 5 places the taxa in mass–length space against lines of constant k, and Figure 6 plots k directly. In every tier the cetacean lies on a higher-k line than the reptile it is compared with. At range midpoints the ratio is 1.3 in tier I, 1.9–2.4 in tier II, 1.4 in tier III and 1.4–2.6 in tier IV, where the lower values use the published regression for the whale.

Two results deserve emphasis. In tier I, the pilot whale reaches the 3–4 t of an 8 m *Tylosaurus* at roughly 5–10% less length. In tier II, the 8.6 m anchor alone ($k = 15.6$) is about twice as full as *Thalassotitan* (8.2 at midpoints), the mosasaurid whose short, broad skull invites comparison with a killer whale's (Longrich et al., 2022).

Tier IV is the least certain. The largest ichthyosaur remains are fragmentary jaw elements from the Rhaetian of the UK: the Lilstock surangular, estimated at about 26 m (Lomax et al., 2018),

Table 4. Body lengths of male killer whales: measured samples, and estimates for unmeasured individuals.

Sample or individual	Method	Typical adult male (m)	Largest male (m)	Status	Source
Japan, coastal catch 1948–57 (320 males)	whalers' reports	6.55 (mean)	9.45 (31 ft)	[P]	Nishiwaki and Handa (1958)
Antarctic type A, Soviet catch 1961–78 (321 whales)	measured on deck	7.3 (mean)	9.0–9.2	[R]	Mikhalev et al. (1981), in Pitman et al. (2007)
Bigg's, British Columbia and Washington 2014–19 (12 adult males)	drone photogrammetry	7.2 (median)	8.3	[P]	Kotik et al. (2023)
North Atlantic type 2	specimens	–	8.5	[P]	Foote et al. (2009)
Southern residents	aerial photogrammetry	6.9 (asymptote)	–	[P]	Kotik et al. (2023)
North Atlantic type 1	specimens	–	6.6	[P]	Foote et al. (2009)
Antarctic type C, Ross Sea 2005 (64 adult males)	aerial photogrammetry	5.6 (mean)	6.1	[P]	Pitman et al. (2007)
CA165, California	field estimate	–	9.1–10.0	[R]	Killer Whale Wiki (n.d.)
Old Thom, Bay of Fundy	press estimate	–	7.6–9.1	[R]	Radio-Canada (n.d.)
John Coe, Hebrides	none published	–	–	[R]	Hebridean Whale and Dolphin Trust (n.d.)

The last three rows are unmeasured individuals. The Soviet maximum is given as 9.0 m in the text and 9.2 m in the abstract of Pitman et al. (2007).

Table 5. Antarctic blue whale females: estimated masses at the lengths discussed.

Position (author's percentiles)	L (m)	Isometric from anchor (t)	With gestational load (t)	Author's bracket (t)	Published regression (t)	Status
45th–65th percentile	27	178	187	–	–	[A]/[T]
Anchor, 1947	27.6	190	199	–	–	[R]
90th–95th percentile	30	244	253	≥240	184–205	[A]/[T]/[R]
99th percentile	32–33	296–325	305–334	270–290	252–273	[A]/[T]/[R]
Longest recorded	33.58	342	351	–	–	[R]/[T]
Theoretical bound	35–37	387–458	–	290–340	–	[T]

Isometric values from equation (1) with $L_0 = 27.6$ m, $M_0 = 190$ t, $G = 1$; they describe a whale as full-bodied as the anchor. The gestational load adds about 9 t. Published regression values are those of Motani and Pyenson (2024) for 30 m and 33 m. Percentile labels are the author's [A] and are discussed in §4.5.

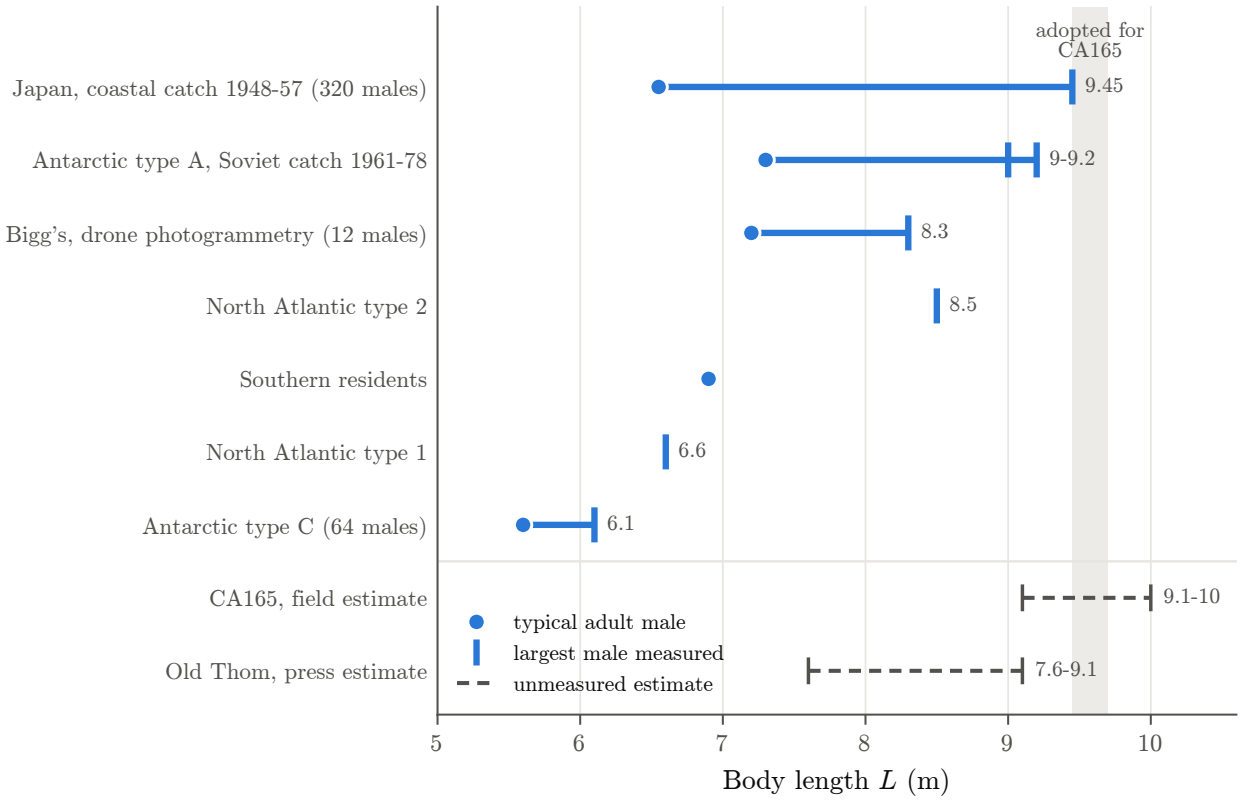


Figure 3. Male body lengths. Blue lines run from the typical adult male (dot) to the largest male (bar) in each measured sample; a dot or bar alone means only that value is published. Grey dashed lines: estimates for unmeasured individuals. Shaded band: the length adopted for CA165 in this paper.

and *Ichthyotitan severnensis*, at about 25 m (Lomax et al., 2024). The Aust Cliff bones have been informally estimated at 30–35 m [R]. We therefore test a 25 m animal and a 36 m upper scenario, for which the author estimates 150–180 t on the assumption of a long tail and a low k [A]. At 25 m with the k of large shastasaurids the animal would weigh 72–90 t [T]; at 36 m with the same, higher k it would weigh 216–268 t [T].

The 25 m animal and the author's 36 m scenario fall below every estimate for a 33 m blue whale. The 36 m scenario with shastasaurid proportions falls below the anchor-condition estimate (325 t) but overlaps the published regression value (252–273 t). At the very top of the size range, then, the ordering depends on the body condition assumed for the whale and the proportions assumed for the ichthyosaur; in the 36 m comparison the whale is the shorter animal.

3.6 Mosasaurid body sizes

Table 7 and Figure 7 summarise commonly cited lengths for representative mosasaurids (Russell, 1967; Longrich et al., 2022). Most described mosasaurids were 3–8 m long, shorter than an adult male killer whale, which is typically 6–8 m and reaches 9–10 m in the largest individuals. Only *M. hoffmannii*, *T. atrox* and *Prognathodon saturator*, together with *T. proriger* on its larger skull-based estimates, reach the 9–12 m range, and none approaches the fullness of a killer whale (§3.5). For *T. proriger* the most complete skeleton measures about 8 m; larger figures rest on isolated large skulls [R].

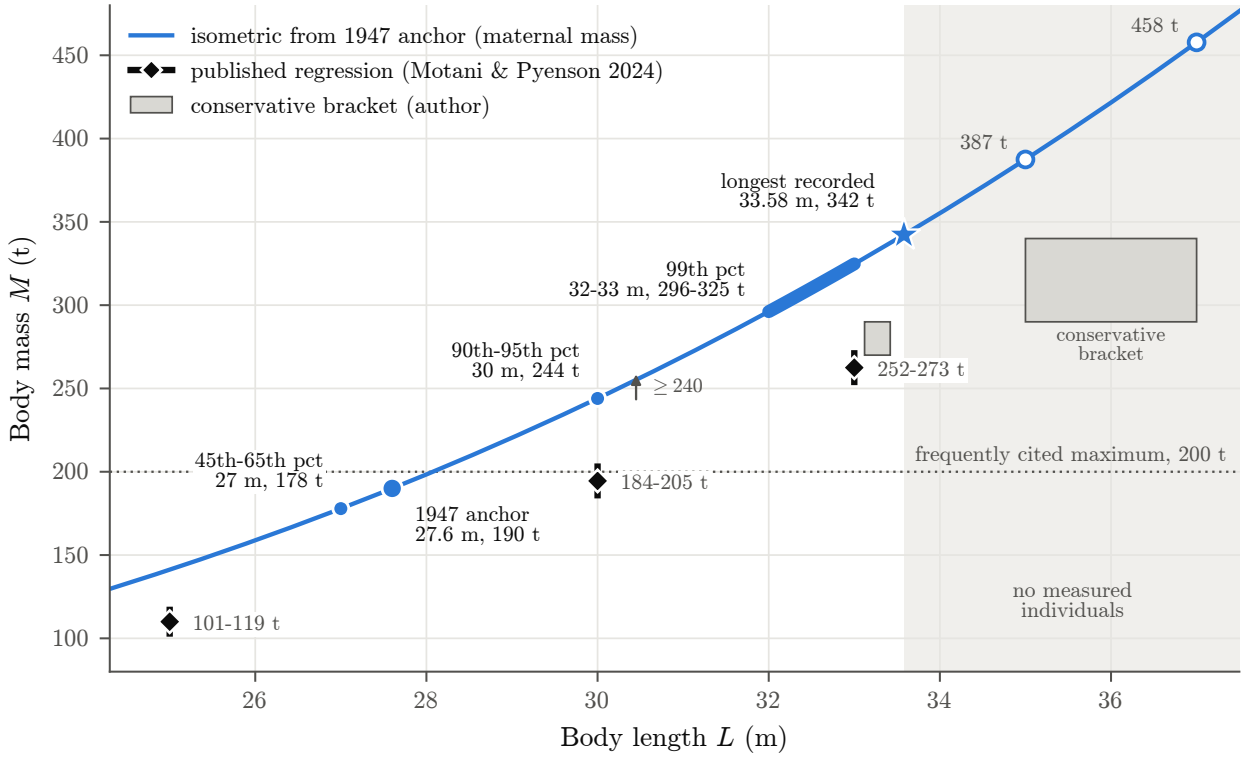


Figure 4. Mass–length estimates for Antarctic blue whales. Line: isometric scaling from the 27.6 m, 190 t maternal anchor, with the author's percentile lengths marked and the longest recorded length as a star. Diamonds with bars: published regression and volumetric estimates (Motani and Pyenson, 2024). Grey boxes: the author's conservative bracket. No individual has been measured in the shaded region beyond 33.58 m. Dotted line: the frequently cited 200 t maximum.

Table 6. Tiered comparison of cetaceans and Mesozoic marine reptiles. $k = M/L^3$ in 10^{-3} t m^{-3}.

Tier	Taxon	Group	L (m)	M (t)	k	Status
I	*Globicephala melas*, large male	Cetacea	7.2–7.6	3–4	6.8–10.7	[R]
I	*Tylosaurus proriger*	Mosasauridae	8.0	3–4	5.9–7.8	[R]/[A]
II	*Orcinus orca*, anchor male	Cetacea	8.6	9.9	15.6	[R]
II	*O. orca*, CA165	Cetacea	9.45–9.70	13.1–17.0	15.6–18.7	[A]/[T]
II	*Thalassotitan atrox*	Mosasauridae	9–10	∼7	7.0–9.6	[A]
II	*Mosasaurus hoffmannii*	Mosasauridae	11–12	8–12	4.6–9.0	[R]
III	*Physeter macrocephalus*, male	Cetacea	16.5	52	11.6	[R]
III	*Himalayasaurus tibetensis*	Ichthyosauria	16.5	35–40	7.8–8.9	[R]/[A]
IV	*Balaenoptera musculus*, anchor	Cetacea	27.6	190	9.0	[R]
IV	*B. musculus*, published regression	Cetacea	30	184–205	6.8–7.6	[R]
IV	Large shastasaurids (*Shonisaurus*, *Shastasaurus*)	Ichthyosauria	20	37–46	4.6–5.8	[R]
IV	Aust Cliff giant, 36 m scenario	Ichthyosauria	36	150–180	3.2–3.9	[A]

k ranges are computed from the ends of the stated ranges; tier ratios in the text use range midpoints. For CA165, k equals the anchor's value times G.

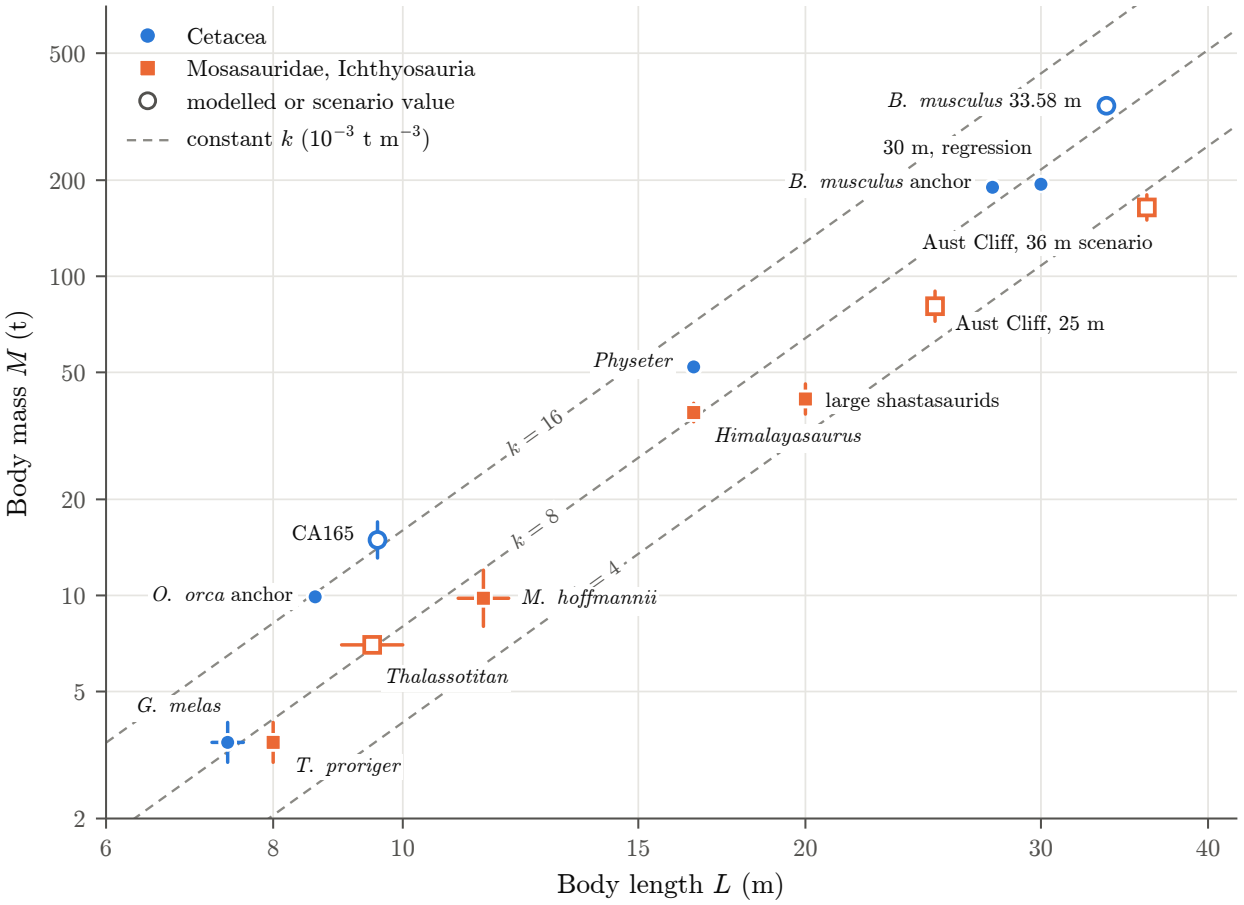


Figure 5. Cetaceans (circles) and Mesozoic marine reptiles (squares) in mass–length space, both axes logarithmic. Bars span stated length and mass ranges. Grey dashed lines are lines of constant fullness k (equation 2); any isometric series runs parallel to them. Open symbols are modelled values [T] or scenarios [A].

Table 7. Representative mosasaurids: commonly cited lengths.

Taxon	Typical adult L (m)	Upper estimates (m)	M (t)	Status
Clidastes spp.	2–4	–	–	[R]
Platecarpus tympaniticus	4–6	–	–	[R]
Khinjaria acuta	∼8	–	–	[R]
Tylosaurus proriger	∼8 (most complete skeleton)	∼13 (isolated skulls)	3–4 at 8 m	[R]/[A]
Thalassotitan atrox	9–10	–	∼7	[R]/[A]
Prognathodon saturator	10–12	–	–	[R]
Mosasaurus hoffmannii	11–12	∼13	8–12	[R]

Lengths are commonly cited values from secondary compilations and should be checked against the primary descriptions before publication.

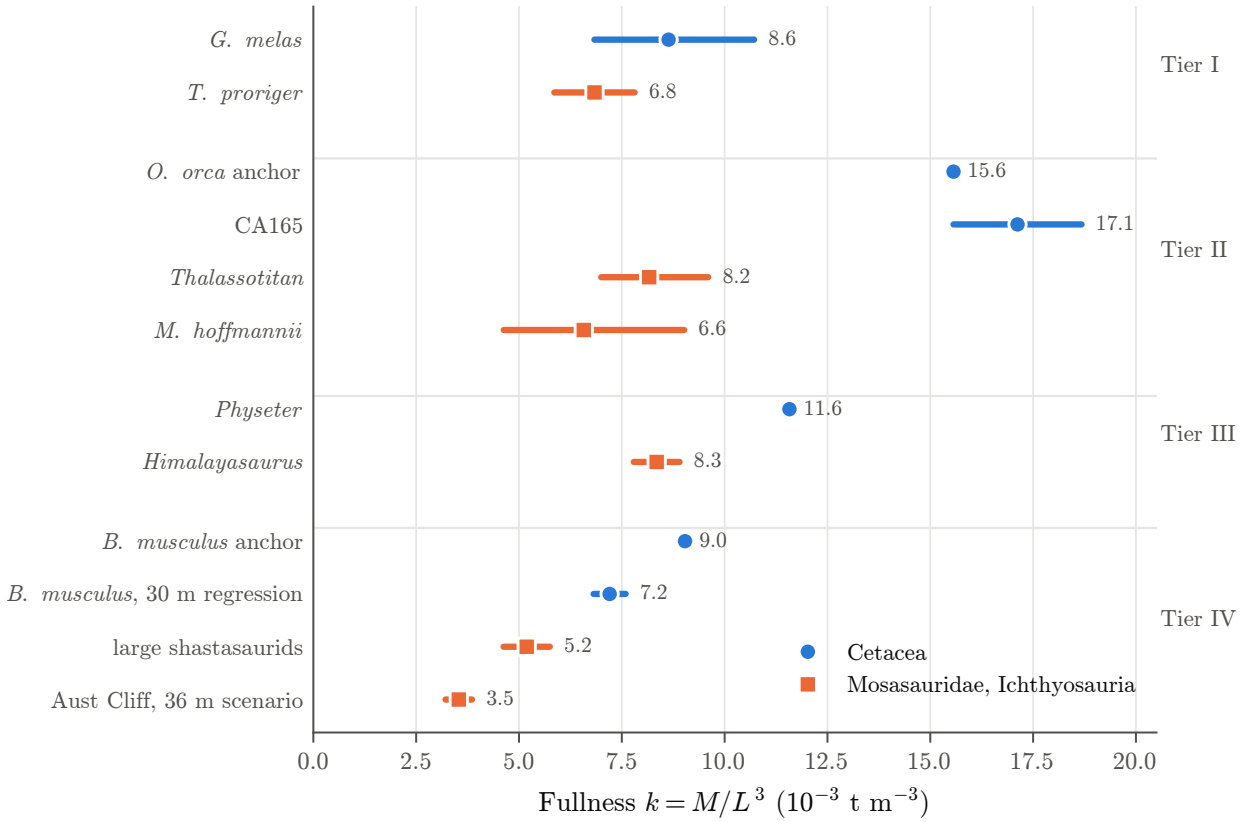


Figure 6. Fullness coefficient $k = M/L^3$ by tier. Lines span the ranges in Table 6; markers are range midpoints. Within each tier the cetacean lies to the right of the reptile.

4 Discussion

4.1 Why cetaceans are fuller

The higher k of cetaceans reflects several features of their construction acting together: a short, deep trunk that carries most of the body volume; a continuous blubber layer, several centimetres thick in killer whales and far thicker in large rorquals, which ectothermic reptiles lack; large locomotor muscle masses sustained by an endothermic metabolism; and a rib cage enclosing large lungs, heart and gut. Mosasaurids, and the large shastasaurid ichthyosaurs, carried a substantial part of their length in a long tail of small cross-section. None of these differences is new. What the fullness coefficient adds is their magnitude: roughly 1.3 to 2.6 times at equal length.

4.2 Sampling the upper tail

Mallon and Hone (2024) modelled a population of 140 million *T. rex* and concluded that the 99th mass percentile has probably been sampled by the 84 known specimens, but that the largest individuals, perhaps 70% heavier than the largest known skeleton, almost certainly have not. The same logic applies with more force to living cetaceans. Far fewer large males have ever been weighed than 84, and weighing is biased toward animals that could be handled whole or processed on a flensing deck. The largest killer whales and blue whales are therefore represented in the literature by estimates, and those estimates have tended to be rounded to familiar numbers rather than derived from the scaling of measured animals.

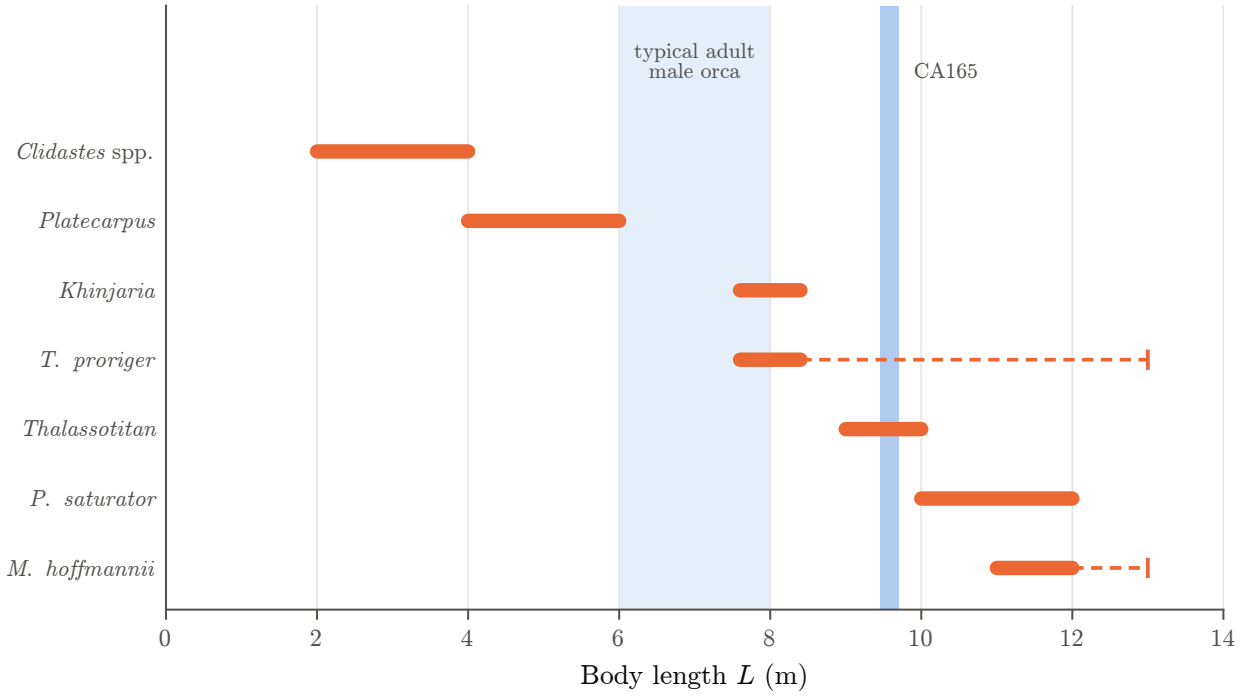


Figure 7. Commonly cited length ranges for representative mosasaurids (bars; dashed extensions are upper estimates from incomplete material), against the typical adult male killer whale (light band, 6–8 m) and CA165 (dark band).

4.3 Convergence of apex predator masses

Table 8 and Figure 8 set the killer whale beside the largest Mesozoic predators. Current estimates for large adult *T. rex* are about 5.4–8.0 t, with about 8.8 t for the largest known specimen and a modelled population maximum near 15 t (Mallon and Hone, 2024). *Spinosaurus* is estimated at about 7.4 t (Sereno et al., 2022), from material that may not be fully grown, and *M. hoffmannii* is usually given at 8–12 t. Large male killer whales, commonly 6–9 t, overlap all of these, and an individual of CA165's length, at 13.4–14.5 t on the Antarctic regression and 13.1–17.0 t on ours, lies above the largest known *T. rex* skeleton and around its modelled maximum. Across three lineages and two media the largest predators converge on the same order of mass, roughly 7–17 t, and the living species is not the smallest of them.

4.4 Asymmetric comparison standards

The familiar picture of Mesozoic marine giants dwarfing modern whales depends on comparing unlike quantities: maximum extrapolated lengths for fossils against population means, or rounded maxima, for living species; and length rather than mass. Held to the same standard, mean against mean or upper tail against upper tail, and measured in mass, most of the gap disappears, and in the upper tiers it reverses.

4.5 Limitations and tests

Each estimate rests on a single anchor, and each anchor carries its own uncertainty; the killer whale anchor in particular must be checked against the original Soviet tables. An anchor in heavy

Table 8. Mass estimates for the largest Mesozoic predators and the killer whale.

Taxon	Group	Estimate	M (t)	Status	Source
Tyrannosaurus rex	Theropoda	large adults, modern estimates	5.4–8.0	[R]	secondary summary
T. rex	Theropoda	largest known specimen (FMNH PR 2081)	≈8.8	[P]	Mallon and Hone (2024)
T. rex	Theropoda	modelled population maximum	≈15	[P]	Mallon and Hone (2024)
Spinosaurus aegyptiacus	Theropoda	volumetric model	≈7.4	[P]	Sereno et al. (2022)
Mosasaurus hoffmannii	Mosasauridae	large adults	8–12	[R]	secondary
Thalassotitan atrox	Mosasauridae	adult	∼7	[A]	this paper
Orcinus orca	Cetacea	large adult males	6–9	[R]	secondary
O. orca	Cetacea	weighed anchor	9.9	[R]	Mikhalev et al. (1981)
O. orca	Cetacea	Antarctic regression, 9.45–9.70 m	13.4–14.5	[R]	Mikhalev (2019)
O. orca, CA165	Cetacea	this study	13.1–17.0	[T]	Table 2

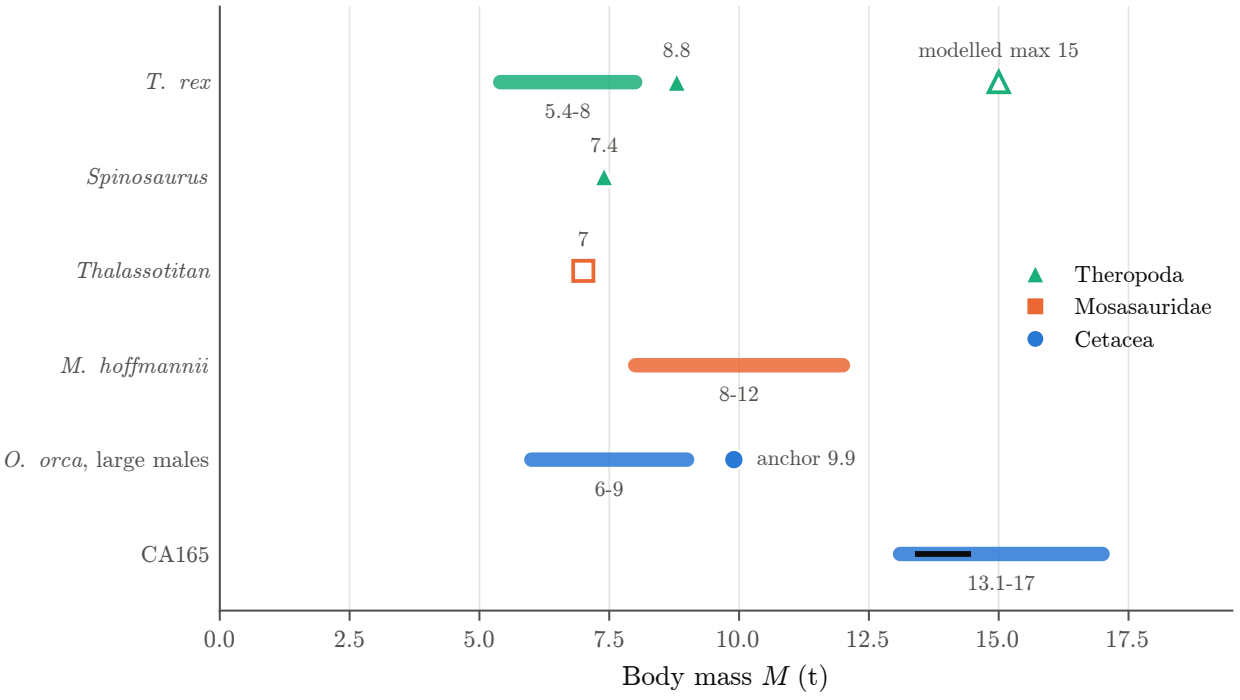


Figure 8. Mass estimates for the largest Mesozoic predators and the killer whale. Bars are ranges; filled markers are point estimates; the open marker is the modelled population maximum for *T. rex* (Mallon and Hone, 2024). The dark bar on the CA165 row is the Antarctic regression at his adopted length (Mikhalev, 2019).

condition raises every value scaled from it. The blue whale anchor demonstrably was (§3.4). The killer whale anchor was not: it lies on the regression for Antarctic whales (§3.1). For the killer whale the larger uncertainties are length (§3.3) and the question whether a relation built on Antarctic whales applies to a Bigg's male. The girth factor for CA165 is a visual judgement. Calibrated drone photogrammetry, of the kind already used to measure free-ranging killer whales (Durban and Parsons, 2006; Fearnbach et al., 2011), would replace it with measured width-to-length ratios and a volumetric mass. Very large rorquals may depart from isometry, which is why the blue whale table carries a conservative bracket; regression and volumetric approaches to rorqual mass are compared by Motani and Pyenson (2024). Fossil masses are reconstructions with uncertainties at least as large as those of the living animals. The percentile positions in Table 5 are the author's. Secondary summaries give a mean length of 25.4–26.3 m for sexually mature Antarctic females and list 88 individuals longer than 30 m in the International Whaling Commission catch database (Wikipedia, 2026) [R]; on those figures 27.6 m lies above the mean, and lengths over 30 m are far rarer than one in ten. The percentiles should be recomputed from the catch-length distribution; the mass scaling does not depend on them. Finally, k depends on how length is defined, and fossil total lengths are not strictly equivalent to notch-to-fluke lengths; the differences involved, a few per cent, are small beside the contrasts reported here.

5 Conclusions

(1) Isometric scaling from a reported 8.6 m, 9.9 t anchor puts a 9.45–9.70 m killer whale at 13.1–14.2 t, and at 14.4–17.0 t with a 10–20% girth excess. Regressions fitted to adult whales give 13.4–15.1 t over the same lengths, so the strict-isometry values are what adult data predict; the girth excess is unmeasured. The "10 t" quoted for a 9.8 m male has no weighing behind it and is about a third too low.

(2) The "200 t" blue whale was a heavy 27.6 m animal, almost 6 m short of the longest recorded. A 33 m animal is estimated at 252–273 t by published regression, at 270–290 t on the author's bracket, and at about 325 t if it carried the anchor's condition.

(3) At equal length, cetaceans are fuller than Mesozoic marine reptiles in every tier examined, by 1.3–2.6 times in $k = M/L^3$.

(4) Most mosasaurids were smaller than an adult male killer whale; a few species reach its length, and none its fullness.

(5) The largest killer whales fall in the same mass band as *T. rex*, *Spinosaurus* and *M. hoffmannii*.

(6) For the killer whale, length is now the main uncertainty. The largest males in measured samples are 9.0–9.45 m, and no individual reputed to reach 9 m, CA165 included, has been measured.

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A Appendix: title and reproducibility

Massa corporis et densitas architecturae, "body mass and the density of construction", states the thesis of the paper: mass, not length, is the proper measure of size, and the construction of the body, summarised here by k, decides how much mass a given length carries.

Every number in the text, tables and figures is computed by `model.py`, which is distributed with the LaTeX source; the figures are drawn by `figures.py` from the same values.